Real-Time Inappropriate Speech Detection

Sai Arya RB
Dept. of CSE-AIML
PES University
Bengaluru, India
saiaryarb2003@gmail.com

Samarth S Kulkarni
Dept. of CSE-AIML
PES University
Bengaluru, India
samarthsk2004@gmail.com

Siddharth Gandhi
Dept. of CSE-AIML
PES University
Bengaluru, India
sidd10.gandhi@gmail.com

Sudarshan Srinivasan
Dept. of CSE-AIML
PES University
Bengaluru, India
strikersuds@gmail.com

Dr. Jayashree R
Professor, Dept. of CSE-AIML
PES University
Bengaluru, India
jayashree@pes.edu

***Abstract*—With the increasing use of online platforms, a conundrum rises on what is inappropriate between different individuals. This dilemma is largely seen in live streaming platforms like Twitch, Kick, YouTube, and various new channels across the world. This inspired our project "Real-Time Inappropriate Speech Detection", with a goal to build a system that can enable platforms to establish a safe environment. The dataset was made using a scraper that finds clips that contain profanity. The data was further cleaned and annotated to ensure that the model processes all the modalities. The model's task is 2-fold – to learn the audio and video features, and then to combine these learned features in order to predict the occurrence of profane content. In particular, we have used a hybrid CNN-LSTM network, which takes Mel-Frequency Cepstral Coefficients (MFCCs) and geometric facial landmarks, and combines them using an Adaptive Late Fusion mechanism. The design will provide a predictive range of 0.5 seconds and allow proactive censorship. The experimental outcomes reveal that the multimodal system reaches a Test F1-score of 0.824 and Recall of 0.859 and is effective in reducing false positives that exist in audio-only systems. Moreover, the system has an inference latency of 0.036 ms per sample, making it computationally efficient and making it quite viable to deploy in real-time live streaming applications.**

***Index Terms*—Multimodal Deep Learning, Profanity Detection, Automated Censorship, Adaptive Fusion, Real-Time Inference**

I. Introduction

Due to recent increase in popularity of Live Streaming platforms, there exists a need for establishing a safe environment for all age demographics by moderating use of profanity that may or may not be intentional. This need is highlighted by incidents on live telecast, such as Dillon Brooks being fined $25,000 for using profane language during a live interview [1], and the penalty of £40,000 on Yuki Tsunoda for swearing during the Austrian Grand Prix [2].

It is difficult to moderate content on live streaming platforms because everything is unscripted. Live streams require continuous monitoring because of its unpredictable nature, which makes it harder to catch inappropriate content such as curse words. It is important to note that real time live streams are not just about one kind of data type but include audio, video, and text all at once. This makes moderation more complex without having a model that can analyze these inputs simultaneously. Foul words need not always be spoken out but can also depend on the body language and facial expression. Along with this, sarcasm and irony can also be difficult to predict only on either spoken words or body language.

A complete representation is difficult to get with traditional single-modal models. Multimodal models can offer a holistic approach by simultaneous analysis of audio, video, and text to overcome the limitations of uni-modal systems. This should improve the accuracy of identifying profanity. It is easier to figure out the full context of what is happening when we combine speech with visual data. By using many modalities of data, the system can get a better understanding and become more reliable. Similarly, only multi-modal systems can notice and predict even the most subtle form of curse words. The model can extract temporal sequences and spatial features, which can be done by using deep learning architectures like Convolutional Neural Network and Recurrent Neural Networks, which can increase efficiency.

We are proposing the use of a multimodal deep learning framework for detecting inappropriate language in real-time systems. The model should focus on relevant features covering different modalities by combining audio features with visual cues and using cross modal attention mechanisms. This will improve its ability to predict inappropriate content. The end goal is to develop a context-aware system that can enable platforms to establish a respectful community.

II. Related Work

The process of determining the modality of data to use was based on 2 main ideas – using individual modalities or to combine them. Smirnov and Laushkina [3] examine the use of only audio as it is faster for real-time profanity detection as compared to ASR, which caused more latency and was less reliable in adverse conditions. Ba Wazir et al. [4] saw that using audio as the only modality was preferred due to its ability to create highly efficient and lightweight systems which are suitable for live use cases. Fayek, Lech, and Cavedon [5] found that it also made preprocessing simple as features can be directly extracted from speech. Lykartsis and Kotti [6] found that speech itself contains verbal signs and emotions which can indicate user opinion. Duncan, Shine, and English [7] focused on video as a modality, finding that many emotions are visually expressed and are more useful when used independently from audio quality, and background noise. Dwijayanti, Iqbal, and Suprapto [8] noticed that these visual cues can also help understand human behavior in facial emotion recognition tasks. Yang et al. [9] used this idea to interpret spoken words based on movement of lips, which is important where audio is not reliable. Serrà et al. [10] found that text is another such modality which can focus on detecting Out of Vocabulary (OOV) words, that consists of offensive words which are intentionally misspelled and new unique slurs. Yang, Grenon-Godbout, and Rabbany [11], and Stoop et al. [12] investigated that profanity is also seen in gaming platforms, with rapid messages being exchanged including slangs related to different games, which needs moderation to try and control the harmful interactions promptly. Moon et al. [13] found that text can also be used to find toxic language, such as hate speech, and norm violations in live streaming platforms like Twitch and YouTube Live. These different points give birth to the idea of multimodality where we combine the features of two or more modalities to build more contextually accurate models. Rana and Jha [14] detected hate speech by combining what is said, i.e., semantics in text, and how it is said, i.e., emotional cues in text. This helps in detecting sarcasm. Bhin, Lim, and Choi [15] proposed that a similar setup can also be used to detect human personality traits such as showing empathy. Pan et al. [16] used these modalities along with brain signals (EEG) to perceive people's emotions better. Chaudhari et al. [17] proposed that multimodality paves the way to build systems that can both detect profanity, using audio, and pixelate content, using video, to maximize censorship. This helps avoid viewers from inferring inappropriate content.

With the strengths and limitations established for each type of data, the logical move is to understand the range of tasks that can be performed. A famous task is the detection and classification of profanity. Some studies find that early intervention is important and do so by mainly using audio features to censor before the full word is spoken [3]. As most existing models are way too bulky for live use, another method used lightweight deep learning models for detecting bad language in speech quickly [4]. In a different view, one study encountered the problem of inferring profane words by sound muting and lip blurring [17]. As the nature of dialect is ever changing, it is crucial that hate speech is detected even in intentionally misspelled words and new unique slurs [10]. Data can also be leveraged to train systems that, in turn, help create better data that can be used to solve more complex downstream tasks, one such case is automated annotations by learning through audio and video analysis. Lenkiewicz et al. [18] found that this cuts down on the time taken to perform the annotations manually and helps increase the uniformity of the annotations, maintaining consistency. The task of recognizing emotions and personalities play a major role in improving the detection and classification task, as it can help identify when someone's getting frustrated or angry right before they swear. One study combines face expressions, speech, and brain signals (EEG) to perceive people's emotions better [16]. Meng et al. [19] found that by using CNNs to capture spatial features along with LSTMs to get the temporal features, there was an improvement in accuracy. They were found to be faster and also maintained accuracy in classification tasks [5]. CNNs are quite capable at real time emotion recognition like happiness, sadness, anger, etc. [7]. Personality is another such trait which can be analyzed to understand more natural response and behavior in different situations [15]. More studies found that context plays a key role in early prevention and intervention of hate speech and messages in chat. This challenge is prevalent in most gaming platforms, where toxic players are detected as the conversation develops, making it possible for gaming companies to intervene by issuing warnings, muting, or banning the player [12]. It is visible that decisions cannot be made by analyzing individual messages, hence analyzing a sequence of messages helps in understanding context [11]. The most interesting task is prediction, which paves way for functional real time systems. Using multimodal data helps in predicting spoken words by analyzing lip movement in videos. Audio is found to carry temporal information, so by transferring this audio knowledge to its corresponding video, the audio-to-lip model generates a video stream that contains the same temporal information. This approach involving cross-modality improved the accuracy of reading lip movement [9].

III. Methodology

A. Data Preparation

The data preparation workflow is a process that acquires raw video data, extracts and preprocesses the audio signals, matches it to a time-aligned transcript, and finds the instances which contain profanity. Initially the video to be processed is downloaded from sources such as YouTube, Twitch etc. The scraper ensures that the configuration is set to the best video and audio streams, before creating a standard MP4 file. From this MP4 file, the audio track is converted into a compatible format which is used for transcription by the model. To ensure that the performance does not hinder, the waveform is segmented into 5-minute chunks, which are exported to temporary WAV files, and then transcribed individually. A word search is performed on these transcriptions to find the timestamps of the predefined profane words. Using these instances, we generate video clips that make the dataset.

The clips in this dataset are then processed to ensure that they are not liable to pronunciation errors, transcription faults, and variations in target words. The clips are further analyzed at the frame-level, where frames containing profanity are labelled as 1 and the remaining as 0. This establishes an alignment between the frames and the audio features for each clip, that is factually correct. In order to build real time systems, we need to focus on prediction rather than classification. This is done using a window to shift the labels, such that profane instances occur after 0.5 seconds of the label. A summary csv file is created to store the metadata of the processed clips.

B. Feature Extraction

We extracted two separate feature streams in order to record both visual and auditory data.

For the audio modality, the audio track of each video was resampled to 16kHz. We then extracted 30-dimensional Mel-Frequency Cepstral Coefficients (MFCCs) using a 10ms hop length. This resulted in a feature matrix of shape (30×T), where T is the number of 10ms frames.

In the case of visual modality, we applied the MediaPipe FaceMesh framework to identify and track 478 facial landmarks. Rather than operating with the raw landmarks, we used temporal spatial correlations to compute a 50-dimensional geometric feature vector, which captures dynamic spatial relationships such as Mouth Aspect Ratio (MAR), Eye Aspect Ratio (EAR), lip distances, jaw angles etc. Video frames were handled at a lower rate of 15 FPS to optimize the performance to real time. This sparse visual feature stream was then temporally interpolated to match the 100 FPS (10ms hop) resolution of the audio features. This resulted in a final, synchronized feature matrix of shape (50×T), where T is the same number of 10ms frames as the audio.

C. Data Sequencing and Augmentation

In order to prepare the data to be used in training, we initially chopped the long stream of features into sequences that were 1.0 second long with a stride of 0.2 seconds, to generate overlapping samples. “Fig. 1”, shows that the most difficult part was that the data is highly skewed, i.e., the majority of clips were clean, and it was extremely rare to find the samples containing swear words. We discovered that normal balancing methods have actually worsened the model and made it over predict profanity on normal speech. This was solved using a multi-part approach:

1) Predictive Labeling: Each of the sequences was assigned a predictive label. We had a view of 0.5 seconds ahead of the window, and in case a swear word appeared in the future segment that we looked at, we scored the whole 1.0 second window as positive.
2) Selective Augmentation: To obtain more of these rare samples of positive data, we applied audio augmentation, i.e., time-stretching, pitch-shifting and adding noise only to the swearing clips. We then made copies of the original face features in order to make matching pairs while maintaining the class balance.
3) Weighted Balancing: We retained all our new positive sequences and reduced the amount of the non-profane class to a smaller ratio , while preserving the natural class distribution by maintaining a higher proportion of the non-profane class , so that our model learns a true prior instead of an artificially imbalanced dataset.. We also trained the model with a weighted loss function to make the model learn on the remaining rare profane samples. This trains the model to be much more careful about profane samples and increase the penalty in case it incorrectly classifies a ”swear” sample.

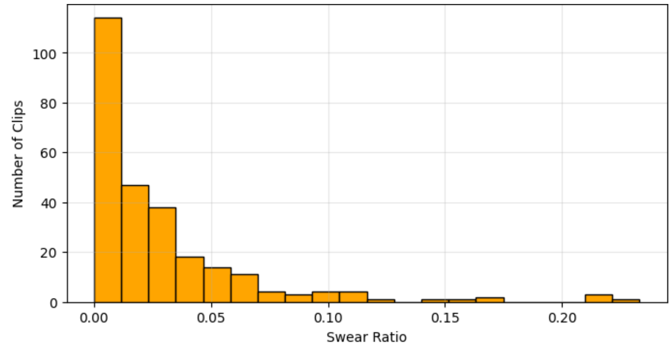

Fig. 1. Swear Frame Ratio per clip

D. Audio Module

The audio module is the most important pillar of the model. The audio model is the primary and most robust component that predicts an upcoming swear word. The audio module balances both high predictive capabilities and low latency in its prediction speed for its usage in real-time scenarios. This component can be broken down into two distinct aspects which are the acoustic feature representation as well as the temporal modeling and the pattern recognition done by the hybrid deep learning architecture.

1) Acoustic Feature Representation: Instead of raw audio waveforms we decided to use Mel-Frequency Cepstral Coefficients (MFCC) as it imitates the non-linear frequency perception of the human auditory system. Thus, it helps the model understand how the human being is perceiving the audio and can learn through that perspective. The multi-step process used in converting the audio to MFCCs involves:

- Resampling and Normalization: All the incoming audio is resampled to a uniform 16000 Hz which is a widely accepted rate for ensuring consistency in the data.
- Framing and Windowing: The signal is then converted into windows of 25ms with an overlap of 10ms. These continuous windows are an indispensable tool which allows us to feed the model the exact starts of words.
- Frequency Domain Transformation: To get a magnitude spectrum, a Fast Fourier Transform (FFT) is done on the signal to convert it from the time domain to a frequency domain.
- Mel Filterbank Application: By applying a bank of triangular filters, the power scale is fit on to the Mel scale. Each filter’s frequency band’s energy is added.
- Logarithmic Compression: We then take a logarithm of the filterbank as the human perception of loudness is logarithmic as well.
- Discrete Cosine Transform (DCT): A Discrete Cosine Transform is performed on the log filterbank energies which gives us our MFCCs. Another thing of note is the energy compaction property of DCTs which result in the lower-order coefficients containing the most vital data.

- Final Parameterization: The process results in a 30-dimension MFCC vector for each 10ms hop. To yield a rich representation without too much complexity, this dimensionality was chosen. For a 1 second clip, we get a final output of a tensor of 30 cepstral coefficients over 100-time steps.

2) Hybrid Deep Learning Architecture: The core architecture contains a Convolutional Neural Network (CNN) for translation-invariant detection and a Long Short-Term Memory for modelling long range temporal dependencies. This helps the model learn the shape as well as sequence of shapes that the signal is producing.

The CNN Architecture:

- First CNN Block: This block processes the 30-dimensional MFCC input with 64 convolutional filters. A kernel size of 3 allows it to capture patterns across three adjacent time steps, while a padding of 1 ensures that the temporal dimension is preserved.
- Second CNN Block: The second block takes the 64-channel output of the first block and applies 128 filters to it, thus learning the higher level feature combinations.
- Stabilization and Regularization:
  - BatchNorm1d (Batch Normalization): Batch Normalization normalizes the activations of the previous layer which combats the problem of internal covariate shifts. This stabilizes the training process, allowing for higher learning rates and provides a slight regularization effect.
  - ReLU (Rectified Linear Unit): ReLU is a non-linear activation function that allows the network to learn complex and non-linear relationships. It is non-saturating in nature and helps mitigate the vanishing gradient problem.
  - Dropout: It is a technique that randomly sets a selection of the neuron activations to zero during training. By being unable to be overly reliant on any one feature, we force our model to generalize on unseen data. Our model uses a dropout rate of 0.2.

The LSTM Architecture:

- Stacked LSTM: A two layer Long Short-Term Memory model was used for hierarchical learning. For example, the first layer may learn acoustic patterns to syllables and the second layer may learn syllables to words or phrases.
- Hidden State Dimensionality: Each LSTM layer has a hidden size 128, which ought to be large enough to represent all the complicated temporal dynamics which occur in a one-second audio file.
- Final Context Vector: The model obtains the final hidden state of the last time-step of the second LSTM-layer which is a single compressed-vector which represents the entire one second audio sequence.

3) The Final Classifier: To project the 128-dimensional output to 64-dimensional space, the final classifier contains a linear layer which is followed by another ReLU activation and a Dropout of 0.3. The training methodology and systemic role involves the following:

- The BCEWithLogitsLoss function is used to optimise the model and a key parameter is the pos_weight which is calculated as the ratio of negative samples to positive samples in the training set. This weight increases the loss incurred when the model misclassifies a positive sample which allows us to fight the bias the model may have towards the majority class and thus improving the model's performance.
- Optimization: Adam is an adaptive learning rate optimization algorithm that has the advantages of both AdaGrad and RMSProp which is used to update our model's weights.

E. Video Module

The Video module is a component that helps the primary audio module to better its prediction by providing it with crucial facial data such as mouth movements, eye movements, emotional expressions, etc. The focus on audio instead of video is to reduce the latency of the model working in real time.

Our system uses a complicated feature engineering method to get around the high cost of processing raw video pixels with deep 2D or 3D CNNs. Using the MediaPipe framework from Google, we turn the video frames into a low-dimensional but high-information geometric feature vector.

This results in dimensionality reduction as well. Raw video frames are extremely high-dimensional. Processing such data directly in real-time is expensive and time-consuming and by comparison, MediaPipe's abstracts each frame into a set of 478 3D landmarks, from which a mere 50-dimensional feature vector is engineered which represents a massive reduction in data complexity. This represents a massive reduction in data complexity. Geometric data is also much more invariant than raw video pixel data.

The feature generation pipeline involves:

- Temporal Down sampling: We only process every other frame in a 30 FPS video, meaning a target rate of 15 frames per second, that halves the computational load which is very important for a model that requires very low latency.
- MediaPipe for Landmark Detection: Each frame that was selected is run though the face_mesh MediaPipe model that identifies a single face and produces a mesh of 478 3-D features that correlate to important parts of the face such as lips, eyes, eyebrows, nose, and jawline.
- Geometric Feature Engineering: The landmark coordinates are used to generate a 50-dimensional feature vector. These capture,
  - Mouth: A majority of the features we try to collect are focused on the mouth such as the Mouth Aspect Ratio, which is the ratio of the vertical opening of the mouth to its horizontal opening.

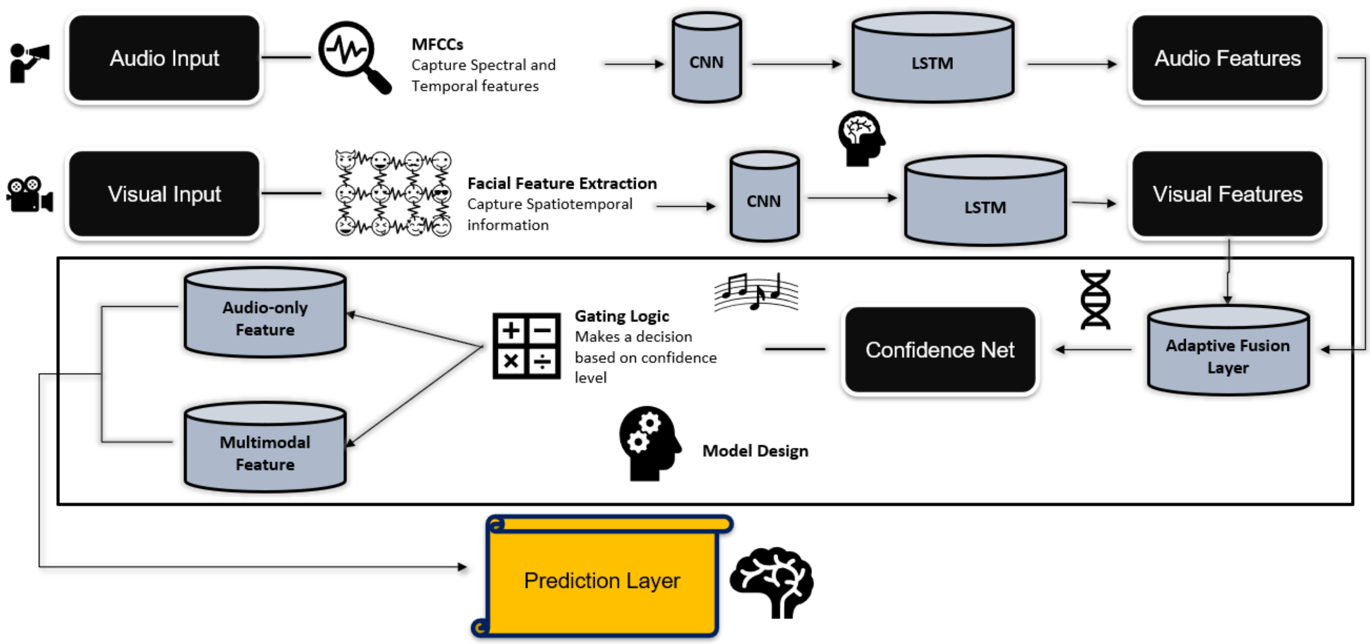

Fig. 2. Model Architecture Design

- Eye: We can use the widening or squinting of the eyes as indicators of emotions and we calculate the Eye Aspect Ratio for both the eyes.
- Eyebrows and Jaw: The eyebrows being raised or the movement of the jaw can give us important features which can help us identify when a profanity is going to be uttered.
- Global and Relational Features: The Face Aspect Ratio is an example of a global feature along with other relationships collected.
- Temporal Synchronization: The feature vectors, which were originally produced at 15 FPS, are on a different time scale as compared to the 100 FPS audio MFCCs. To integrate these modalities, there has to be a complete alignment. Linear interpolation is used to achieve this. The 15 FPS sequence of visual features is up sampled to 100 FPS creating new feature vectors for the intermediate time steps. The weighted average of the values from the two nearest original frames is used as the value of the new feature. Every 10ms audio feature now has a corresponding time-aligned video feature.

1) Lightweight CNN-LSTM Architecture: The neural network architecture for the visual model is designed to be around the same as the audio model, employing a CNN-LSTM hybrid. However, it is intentionally designed to be significantly lighter and computationally cheaper.

The CNN layer architecture:

- Functionality: Similar to the audio model, the Conv1d layers detect local, time-invariant patterns in the sequence of geometric features. Here, they learn to recognize short, characteristic facial movements.
- Architectural Differences: The visual CNN is shallower and narrower than its audio counterpart.
  - First CNN Block: Processes the 50-dimensional input with only 32 filters.
  - Second CNN Block: Expands the representation to 64 feature maps. This is a marked reduction compared to the audio model's 64 to 128 filter structure, leading to a substantial decrease in the number of parameters.

The LSTM layer architecture:

- Functionality: To process the sequence of feature maps to model the change in facial gestures over time, we use an LSTM model.
- Architectural Differences: Instead of using a 2-layer architecture, like in the audio model, we use a model with only a single layer.

2) The Final Classifier: The Final Classifier, i.e., fully connected layers includes a classifier head that is also streamlined, with a smaller number of neurons in its hidden layer, further contributing to the model's overall efficiency.

F. Inference and Censoring Pipeline

In order to implement the trained model, we constructed an inference pipeline that processes videos that have never been seen. "Fig. 2" shows that the pipeline initially extracts and synchronizes audio (MFCC) and facial (geometric) features, as done in training. It then processes these features using the model with the same 1.0-second sliding window in order to produce a continuous probability stream.

We use a temporal median filter to smooth these predictions so as to reduce false positives. The smoothed probability is compared against the threshold, and a "bleep" is scheduled into the system when the probability is greater than the threshold value. Since the model was trained with a 0.5-second prediction horizon, the model is capable of muting the audio before the swear word is said. Lastly, the system cuts the original video and rejoins it with this now bleeped audio track to produce the final censored video.

IV. Results

This section shows the quantitative and qualitative findings of the model training and assessment. The experiments were aimed at determining the high-performance audio-only state, testing the functioning of the final multimodal system, and determining the possibility of using the system in real-time.

A. Experimental Setup

The augmented and sequenced data were 58,404 sequences, each one second long. The data set was divided into training (46,723 samples), validation (5,840 samples) and test (5,841 samples) sets, with the stratified positive class (profanity) ratio of 16.7%.

All of the models have been trained on a CUDA-enabled GPU. In order to cope with the large amount of class imbalance, pos_weight of 5.0 was added to the BCEWithLogitsLoss function to add the penalty on the misclassifications of the rare positive class.

B. Audio-only Features Performance

An audio-only CNN-LSTM baseline model was trained to create a performance baseline. The model showed high discriminative ability as it had a Test AUC of 0.959. Once the decision threshold on the validation set was optimized to 0.873, an increased result over the default of 0.5, the model reached a final Test F1-score of 0.826. Table 1 shows that the model had a high recall of 0.823 on the profanity class.

C. Multimodal Features Performance

One of the main peculiarities of the dataset was that only 7.0 per cent of the training sequences included any facial features that could be detected, which restricted the data supplied to the visual branch.

The final multimodal model had a Test AUC of 0.960 and Test F1-score of 0.824, at optimal threshold of 0.826 which can be seen in "Fig. 3". The results of this model, as revealed by Table 1, yielded a statistically significant improvement in recall of profanity to 0.859, which identified a greater number of true positive cases than the baseline, although with a little less precision.

TABLE I
Performance Comparison on Test Set

Model	F1-Score	Test AUC	Recall	Precision
Audio-Only (Baseline)	0.826	0.959	0.823	0.829
Multimodal (Fused)	0.824	0.960	0.859	0.792

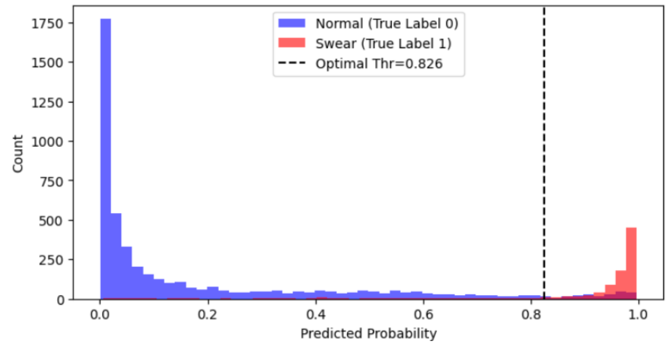

Fig. 3. Prediction Probability Distribution by True Class

D. Comparative Analysis

When the two models are compared directly, it is seen that final F1-scores were close to each other. The behavior of the multimodal model however changed to favor higher recall.

The trained AdaptiveFusionLayer was analyzed and it was found that it had learned to give a weight of 0.90 to the audio stream and a weight of 0.06 to the visual stream. This means that the fusion model correctly identified that the audio stream is to be used as the main and most accurate predictor due to the lack of visual data. Effectively, the adaptive model was able to learn how to revert to a superb audio-centric model and use the available limited visual data in order to improve recall.

E. Beyond the Audio: Facial Cues to increase the accuracy of profanity recognition

While testing on a clip extracted from a stream, consisting of 2 swear words - one at 3 secs and the other at around 6 secs, the audio specific model predicts numerous profane occurrences as shown in "Fig. 4" , while the final multimodal model correctly eliminates false positives as shown in "Fig. 5".

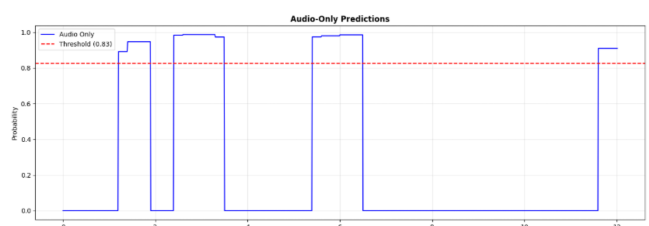

Fig. 4. Audio-Only Prediction on a live sample

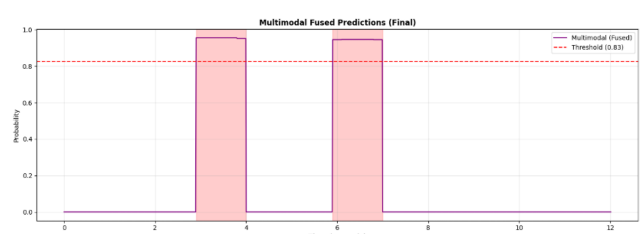

Fig. 5. Final Multimodal Prediction on a live sample

F. Latency and Real-Time Feasibility

A latency test was carried out to test the validity of the system to be used in real-time. The complete multimodal model proved to be very efficient with an achievement of:

- Average Latency per Sample: 0.036 ms
- Throughput: 27,688 samples/second

These findings ensure that the computational cost of the model is insignificant and has the potential of executing in a real-time, or faster-than-real-time, processing environment.

V. Conclusions

In this paper, we explored the notion of real-time profanity prediction in live streams and demonstrated how a multimodal model (Audio and Video) can achieve this feat with high accuracy and low latency. Through systematic analysis, we showed that our system can process audio and video streams and is suitable for real-time deployment in practical scenarios.

Overall, our work highlights the potential of a lightweight architecture for real-time profanity detection.

Our model can be improved to consider a higher weightage for facial features, lower false positives and improve latency. This can be achieved by training on a larger amount of data. Latency is real-time but currently in the context of a GPU environment, this can be reworked to function in any environment regardless of hardware.

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